

What if designing a bathroom felt like playing with a real, physical space - instead of scrolling through a catalogue?

This project takes a room's dimensions, a budget, and a style in plain language, and turns them into a fully furnished 3D bathroom - fixtures picked, placed, and rendered in real time, ready to walk through and tweak on the spot.

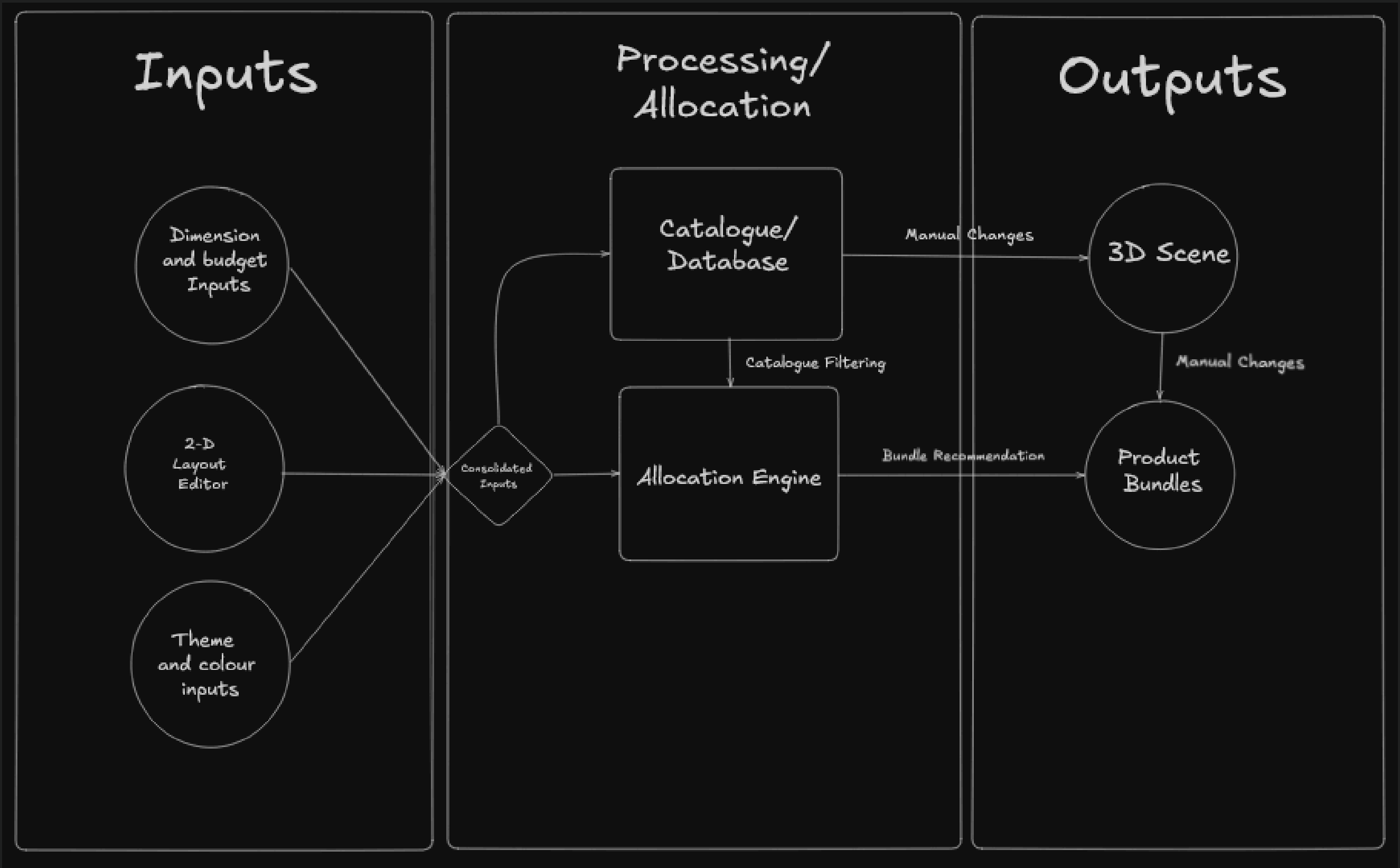


Tech Stack

Layer	Technology	Purpose
Backend framework	FastAPI	REST API server
AI / LLM	Gemini API	Bundle matching, catalogue enrichment, allocation engine
Frontend framework	React (Vite)	UI & app shell
3D rendering	React Three Fiber (@react-three/fiber) + drei (@react-three/drei)	3D bathroom scene, OBJ model loading, camera control
2D planner	Custom React component (LayoutPlanner2D.jsx)	Drag/arrange fixtures, budget & spec inputs
Asset format	.obj files	3D fixture models



Workflow





Features and Innovations

One-click allocation — A single prompt generates a complete, budget-matched fixture bundle instantly.

Dynamic camera movement — The 3D camera smoothly glides to and from any fixture you select, then returns exactly where it left off.

Gamified interactions — Click a fixture to inspect it up close, click off to zoom back out — designed to feel like exploring a space, not browsing a spreadsheet. Also allows for cycling between a product type with ease, to see how it fits with your bathroom.

Real-time visualization changes — Every edit - theme, product swap, layout tweak - reflects instantly in the live 3D scene.

Detailed product catalogue — Browse real 3D-rendered thumbnails with finishes, dimensions, and tags instead of static stock images.

Ease in product selection — Swap any AI-picked item manually, with clear "already added" indicators by SKU.

Modular 2D editor — Drag, arrange, and reconfigure products and zones (shower, tub, fixtures) freely, with automatic collision and fit checks